

Collapse Lemmas — Euler Equations

Lemma 1: Boundedness of Scaffold Field $\psi(x, t)$

Statement: Let ψ evolve under

$$\partial_t \psi + (u \cdot \nabla) \psi = -\alpha \psi + \nabla \cdot \mathcal{T}(u),$$

with $\alpha > 0$, $u \in H^s(\mathbb{R}^3)$, $s > \frac{5}{2}$, and $\mathcal{T}(u) \in L^2$. Then $\|\psi(t)\|_{L^2} \leq C$ for all time.

Proof Sketch: Take L^2 inner product with ψ , integrate:

$$\frac{1}{2} \frac{d}{dt} \|\psi\|^2 + \alpha \|\psi\|^2 = \int \nabla \cdot \mathcal{T}(u) \cdot \psi \, dx$$

Right-hand side bounded by $\|\mathcal{T}(u)\| \cdot \|\nabla \psi\|$, closed via Grönwall.

Lemma 2: Preservation of Inviscid Energy Structure

Statement: Define symbolic total energy:

$$E(t) = \int \frac{1}{2} |u|^2 + \lambda |\psi|^2 \, dx$$

Then $E(t)$ remains finite for all $t > 0$.

Proof Sketch: Euler part conserves $\int |u|^2$, ψ contribution dissipates via Lemma 1. The symbolic structure acts as **nonlinear regulator** without breaking the inviscid conservation law.

Lemma 3: Suppression of Gradient Blowup

Statement: There exists $\delta > 0$ such that:

$$\|\nabla u(t)\|_\infty \leq C + \delta \|\psi(t)\|_{H^1}$$

Proof Sketch: Use vorticity equation:

$$\partial_t \omega + (u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nabla \times \psi$$

$\nabla \times \psi$ suppresses the vorticity stretching term, taming the feedback loop. Control maintained by H^1 boundedness of ψ .

Lemma 4: Classical Limit Preservation

Statement: As $\psi \rightarrow 0$ in L^2 , solutions $u \rightarrow u_E$ satisfy the classical Euler equations and retain smoothness if initial data is smooth.

Proof Sketch: - Weak limit passes through due to uniform bounds - No viscosity term needed for compactness - Regularity transferred via symbolic scaffold control

These lemmas establish recursive regularity in Euler equations without dissipative forces, relying purely on symbolic collapse feedback.